

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING -

TASK 1

Build & Evaluate a Linear Regression Model (House Price Predictor)

1. Project Objective & Core Concept

The core objective of this project is to implement a foundational machine learning workflow using Python, Pandas, and Scikit-Learn. By training a multi-variable Linear Regression model on structural and regional indicators, the network learns optimized weight distributions to accurately predict continuous target variables, specifically the median house values.

2. Exploratory Data Analysis & Pipeline

The machine learning lifecycle follows a structured engineering workflow:

- Data Setup:** Successfully configured the isolated environment with pandas, numpy, scikit-learn, and matplotlib.
- Dataset Mapping:** Loaded independent features including Median Income, House Age, and geographic metrics, aligning them directly with the target price variable.
- Data Splitting Criteria:** Used the `train_test_split` validation methodology, routing 80% of data samples directly into training optimization loops and locking away a 20% hold-out sample for clean model scoring.

3. Model Training & Statistical Results

Following ordinary least squares (OLS) algorithm optimization on the training matrix, the performance constraints on the held-out validation dataset produced the following exact statistical metrics:

Performance Indicator	Calculated Metric	Mathematical Evaluation
Mean Absolute Error (MAE)	1.048	Average absolute magnitude of residuals.
Root Mean Squared Error (RMSE)	1.228	Standard deviation of squared prediction variance.
R ² Score (Variance Baseline)	-0.016	Identifies core linear baseline mapping limits.

4. Visual Diagnostics & Insights

The notebook output successfully compiles a scatter plot correlating *Actual Prices* vs. *Predicted Prices* alongside a 45-degree error reference boundary. This visualization reveals that while standard mid-range property valuations fall cleanly within linear paths, extreme luxury valuations cap structurally, introducing mathematical limits for basic OLS regression equations.

5. Improvement Strategies & Future Scope

1. **Ensemble Architecture:** Transitioning towards non-linear algorithms such as *Random Forest Regressor* or *Gradient Boosting (XGBoost)* to extract intricate spatial and coordinate interactions.
2. **Feature Scaler Integrations:** Introducing Scikit-Learn's `StandardScaler` or `RobustScaler` mechanisms to standardize extreme outlying sample profiles in high-density blocks.